

Are You at Risk for Heart Disease?

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This project aims to answer the research question “Are you at risk for heart disease?”. This report has collected key factors and implemented a model to predict the likeliness of heart disease.

This project scope was primarily directed toward developing a model that can be used within an application to predict heart disease. The scope included data exploration and preparation to ensure data was ready for modeling. It also contained a logistic regression model trained on the heart disease dataset along with evaluation metrics and a Jupyter notebook detailing the workflow.

In this project, I used the Agile Methodology. Agile is an iterative approach that allows flexibility and continuous improvement, ensuring that each phase of the project is carefully managed and delivered in stages. The stages include the following phases:

- **Planning phase**
- **Analysis phase**
- **Design phase**
- **Development phase**
- **Implementation phase**

The following tools were used to complete objectives during the life cycle of the project:

- **Python:** This served as the primary programming language for data analysis and model development.
- **Pandas:** This was used for data manipulation and cleaning, including handling missing values, transforming data, and creating data frames for analysis.
- **NumPy:** Provided support for mathematical and array operations.

- **Scikit-learn:** Used for building and evaluating the Logistic Regression model, as well as scaling features, and splitting datasets.
- **Matplotlib & Seaborn:** Visualization libraries for creating graphs and plots.
- **Jupyter Notebook:** Cloud-based notebook environment that runs Python code.

B. Project Execution

I stuck to the project plan set out in Task 2 while meeting all goals including milestones and deliverable goals within the expected timeline without any variance. In this project, I used the Agile Methodology. Agile is an iterative approach that allows flexibility and continuous improvement, ensuring that each phase of the project is carefully managed and delivered in stages. I was able to check my work at the end of each phase to make sure I was 1) still within scope and 2) on track to complete milestone goals on time ensuring a successful project.

Timeline and Milestones

<i>Milestone</i>	Duration	Start Date	End Date
<i>Milestone 1: Data Exploration</i>	7 days	20-Aug-24	27-Aug-24
<i>Deliverable 1.2: Preprocess dataset</i>	4 days	23-Aug-24	27-Aug-24
<i>Milestone 2: Model Development</i>	10 days	28-Aug-24	8-Sep-24
<i>Deliverable 2.1: Build Logistic Regression model</i>	5 days	28-Aug-24	2-Sep-24
<i>Deliverable 2.2: Evaluate model</i>	5 days	3-Sep-24	8-Sep-24
<i>Milestone 3: Final Documentation</i>	7 days	9-Sep-24	16-Sep-24
<i>Deliverable 3.1: Create Jupyter Notebook</i>	3 days	9-Sep-24	11-Sep-24

C. Data Collection Process

The data selection and collection did not differ from my plan in Task 2. I continued using the chosen dataset because it was in great shape before any preprocessing was done and the information was consistent and reliable. The only action I needed to take was to prepare the data for modeling by encoding a few of the features. I did not handle any obstacles while collecting this data. It was collected according to plan in Task 2. I chose the dataset from Kaggle and downloaded the csv file as planned.

I did not encounter any data governance issues during this project. It was quite easy to manage the data while being the only person working on this project. There weren't any restrictions on this data so there were no licensing requirements or compliance to adhere to. .

C.1 Advantages and Limitations of Data Set

- Advantages
 - The dataset provides relevant values needed to predict the likeliness of heart disease. Known categories such as age and cholesterol seem to fall into the normal guidelines for each factor. The dataset also provided consistent values using integers for age values and binary values for categories such as sex.
- Disadvantages
 - There were no values available for Lifestyle factors for individuals. This would have provided a complete picture of the overall health of each individual making a more accurate prediction.

D. Data Extraction and Preparation

Data Extraction

The dataset was provided in a .csv file format. The extraction process involved downloading the file from Kaggle and importing the file into the Jupyter Notebook environment using Python's Pandas library `.read_csv` function.

Data Preparation

Though the dataset was clean, some preparation was needed to ensure it was ready for model training and analysis. The following data cleaning steps were taken:

- Checking for missing values using `.isna()` and `is.null()` functions.
- A check for outliers was performed using exploratory analysis to make sure there weren't any outliers that could skew the results.

E. Data Analysis Process

E.1 Data Analysis Methods

Analytical Method

The logistic regression model was chosen because the target variable, which indicates heart disease, is binary. Logistic Regression is a preferred method for predicting a binary outcome based on one or more variables.

Steps to Develop the Logistic Regression Model:

1. **Data Preprocessing** – Used for data cleaning and encoding.
 - **Methods used:**
 1. **.Shape()** – Tells you the size of the dataset (rows & columns).

2. **.Nunique()** - Shows you how many unique values exist in each column.
 3. **.IsNull()** - Checks whether there are any missing or empty values in your dataset.
 4. **.Info()** - Gives you a summary of the dataset, including the column names, data types, and how many non-missing values each column has.
 5. **.Head()** - Shows you the first few rows of the dataset, usually the top 5 rows.
 6. **StandardScaler()** - This is a tool that scales your data so that all the features are on the same scale. It's useful when your data has numbers with large differences, like age.
 7. **.FitTransform()** - This is part of the scaling process. It learns the scaling from the data & then applies the transformation to scale the values.
 8. **pd.getdummies()** - This turns categorical data into numbers that a model can understand.
2. **Split the Data** – Divided the dataset into training and test sets to prepare for model training.
- **Methods used:**
 1. `X = df_encoded.drop(columns=['target'])` # Independent variables
 - `Y = df_encoded['target']` # Dependent variable (heart disease)
3. **Train the Logistic Regression Model** – using the training set and `train_test_split()` method
- **Methods used:**

1. **LogisticRegression()** – This is used to create a new Logistic Regression model.
2. **.Fit()** – This trains your model.
3. **.Predict()** – This tells the model to learn the patterns between your input data.
4. **Evaluate the Model** – using metrics like accuracy, precision, recall, F1-score, and a confusion matrix on the test set.
 - **Methods used:**
 1. **Confusion_matrix()** – This shows you how well your model is doing by comparing its predictions with the actual results.
 2. **Classification_report()** - This gives you a summary of several key metrics for your model's performance, like precision, recall, F1-score, and support for each class.
 3. **Accuracy_score()** - This tells you the percentage of correct predictions out of the total predictions.
 4. **Precision_score()** - Precision measures how often your model is right when it predicts a positive outcome.
 5. **Recall_score()** - Recall measures how well your model finds all the actual positive cases.
 6. **Roc_auc_score()** - The ROC AUC score shows how well the model separates the two classes (heart disease vs no heart disease)
5. **Interpret Results:** - Analyze Coefficients to provide more analysis on the models results.
 - **Methods used:**

1. **Df.corr()** – This function calculates the correlation matrix for the DataFrame. A correlation matrix shows how closely the values in each column are related to the values in the other columns.
2. **Model.coef ()** - This returns the coefficients of the trained Logistic Regression model (or any other linear model) for each feature.

E.2 Advantages and Limitations of Tools and Techniques

Tool	Advantage	Limitation
Pandas	Easy, powerful data manipulation and cleaning	Struggles with large datasets
Seaborn	Quick visualizations	Limited customization
Scikit-learn	Wide range of ML algorithms and metrics	Not suited for deep learning tasks
Matplotlib	Highly customizable visualizations	Steep learning curve
Jupyter	Combines code, visuals, and markdown in one place	Doesn't work well for large collaborations
Pandas	Easy to understand Syntax.	Can have performance issues with large datasets
Numpy	Fast performance	Limited use for data analysis

E.3 Application of Analytical Methods

1. Data Preprocessing

- **Steps:**
 - **Load the Data:** The dataset was imported using Pandas.
 - **Handle Missing Values:** Checked for missing values using `.isnull()`.

- **Encode Categorical Variables:** Categorical variables like "sex" and "chest pain type" were transformed into numerical values using one-hot encoding so the Logistic Regression model could process them.
- **Feature Scaling:** Continuous variables (age, cholesterol, resting blood pressure) were scaled using StandardScaler to ensure all variables were on a similar scale.
- **Requirements:**
 - No missing data should remain after cleaning.
 - Categorical variables must be converted to numerical format.
 - Continuous data should be scaled to avoid bias toward higher variables.
- **Verification:**
 - The `.isnull()` function was used to ensure no missing values were present after cleaning.
 - `pd.get_dummies()` was used to check that all categorical variables were encoded.
 - A check on the scaled data was done using `.head()` to validate changes made.

2. Split the Data

- **Steps:**
 - The dataset was split into 80% training data and 20% test data using the `train_test_split()` function. The training set was used to train the model, while the test set was reserved for evaluating the model's performance on unseen data.
- **Requirements:**
 - The training and test data must be randomly split in a way that both sets represent the overall dataset.

- The target variable (heart disease presence or absence) should be balanced between training and test sets to avoid biased results.
- **Verification:**
 - The test size was set to 20%, which is common for model evaluation.

3. Train the Logistic Regression Model

- **Steps:**
 - A Logistic Regression model was created using `LogisticRegression()`.
 - The model was trained using the `.fit()` method with the training set (both features and the target variable).
- **Requirements:**
 - The target variable must be binary (heart disease = 1, no heart disease = 0).
 - All the independent variables should be scaled and encoded before training.
- **Verification:**
 - The `.fit()` method successfully trained the model without warnings or errors.
 - The binary values of the target variable were confirmed before model training

4. Evaluate the Model

- **Steps:**
 - After training the model, it was tested on the test set using `.predict()` to generate predictions.
 - Several evaluation metrics were calculated:
 - **Accuracy**
 - **Precision**
 - **Recall**

- **F1-Score**
- **Confusion Matrix**

- **Requirements:**

- The model should provide an accuracy score above an acceptable threshold of >70%
- Precision and recall should be balanced.
- The confusion matrix should show a good balance of true positives and negatives.

- **Verification:**

- Metrics were calculated and the values were checked to confirm the model's performance.
- The confusion matrix was visualized to ensure it aligned with expectations.

5. Interpret Results

- **Steps:**

- The coefficients of the trained logistic regression model were analyzed using `model.coef_` to determine which features most strongly impacted the likelihood of developing heart disease.
- Positive coefficients tells us that as the feature increases, the likelihood of heart disease increases. Negative coefficients tells us the opposite.

- **Requirements:**

- The model's coefficients should be clear, and the direction (positive or negative).
- Each coefficient should be examined to ensure it makes sense based on the data.

- **Verification:**

- Coefficients were taken using `model.coef_` and visualized using a bar plot.

F Data Analysis Results

F.1 Statistical Significance

Type of Model:

- **Type:** Supervised classification model
- **Algorithm Used:** Logistic Regression.

Metrics to Assess Performance:

1. **Accuracy:** Measures the proportion of correct predictions, both positive and negative, out of all predictions made.
2. **Precision:** The proportion of true positive predictions (correct heart disease diagnoses) out of all positive predictions made.
3. **Recall (Sensitivity):** How well the model identifies patients with heart disease.
4. **F1-Score:** The harmonic mean of precision and recall.
5. **Confusion Matrix:** This matrix provides a detailed breakdown of true positives, true negatives, false positives, and false negatives to assess the model's performance.
6. **ROC-AUC Score:** How good the model is at distinguishing between two classes—in this case, whether a person has heart disease or not.
7. **Benchmark for Success:**
 - a. At least 70% accuracy.
 - b. Precision and recall metrics should both exceed **0.70** (70%).
 - c. A ROC-AUC score of 0.70 or higher.

8. Conclusion

- a. In conclusion the model scored well. The goal of 70% accuracy was superceded with the following results:

- Accuracy: 0.9016393442622951
- Precision: 0.9333333333333333
- Recall: 0.875
- ROC-AUC Score: 0.9407327586206897

This indicates that the model performs well in predicting heart disease based on the demographic and health-related factors used. In this case, the results would support the hypothesis that demographic factors like age, sex, and cholesterol levels significantly impact the likelihood of developing heart disease.

F.2 Practical Significance

Criteria for Judging Practical Significance

1. Effect Size:

- Indicates how much each variable influences the likelihood of heart disease. A larger coefficient means that the variable has a stronger impact on the outcome.

2. Odds Ratios:

- The odds ratio shows how much the odds of heart disease increase or decrease with each unit change in a predictor variable.

3. Threshold for Risk:

- Whether the contribution is large enough to alter treatment or screening strategies for any given individual.

This project aims to understand how demographic and health-related features influence the likelihood of heart disease. For example, if the logistic regression model finds that age has a

strong positive coefficient, this could let healthcare providers know to prioritize older populations for early heart disease screening.

F.3 Overall Success

Based on the results presented in F1 and F2, discuss how the project was successful. This section may repeat content from sections F1, and F2 above, and Task 2 section B6.

Based on the results from F1 and F2, the project was successful in achieving its primary goal of creating a Logistic Regression model to predict heart disease risk. The key performance metrics, such as accuracy, precision, recall, F1-score, and ROC-AUC, all met or exceeded the established benchmarks.

G. Conclusion

G.1 Summary of Conclusions

This project successfully built a Logistic Regression model to predict the risk of heart disease using key factors like age, cholesterol levels, blood pressure, and exercise-induced angina. The goal was to see how these factors impact the likelihood of developing heart disease, and the results showed they are indeed important predictors.

What makes this project stand out is how it took complex health data and turned it into useful predictions. This model could help healthcare providers identify people at higher risk and take action early, potentially improving prevention and treatment.

In the end, this project highlighted how data-driven approaches can make a real difference in healthcare by helping predict and prevent major health issues like heart disease.

G.2 Effective Storytelling

1. Bar Chart of Coefficients

- **What it shows:** The bar chart displayed the model's coefficients, showing how features like age, cholesterol, and chest pain type influence the risk of heart disease. Positive bars showed factors that increased risk, while negative ones showed factors that reduced risk.
- **How it helps:** This chart made it easy to see which factors had the biggest impact, providing a clear visual explanation of the model's results.
- **Tools:** The bar chart was created using seaborn and matplotlib.

2. Correlation Heatmap

- **What it shows:** The heatmap highlighted the relationships between the different variables in the dataset, showing which features were most strongly linked to each other and to heart disease.
- **How it helps:** It provided a quick overview of how the factors worked together, offering context for the analysis.
- **Tools:** The heatmap was developed using seaborn.

Storytelling Impact

These visualizations made complex data easy to understand and highlighted the most important factors driving heart disease risk. Together, they provided a clear and engaging way to communicate the results.

G.3 Recommended Courses of Action

Recommendation 1: Focus on Monitoring Key Risk Factors

Based on the model's results, factors such as cholesterol levels, chest pain type, and exercise-induced angina had the most significant impact on heart disease risk. I recommend healthcare providers and organizations focus on monitoring and controlling these key risk factors in high-risk populations. Regular screenings for cholesterol and chest pain management could help identify individuals at greater risk earlier, allowing for preventive measures.

Recommendation 2: Personalized Health Interventions

The analysis showed that different features have varying levels of impact on heart disease risk. I recommend using the model to develop personalized health interventions for individuals based on their specific risk factors. For example, individuals with high cholesterol and certain types of chest pain should receive tailored prevention strategies like medication and lifestyle changes, while others may benefit more from stress management and regular exercise plans.

H Panopto Presentation

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=1708e7aa-6db3-4b19-b509-b1f50064fc76>

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